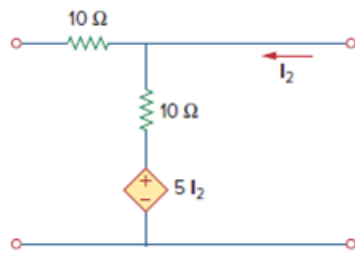


## Problem

Find the  $y$  parameters for the circuit in Fig. 19.81.

**Figure 19.81**

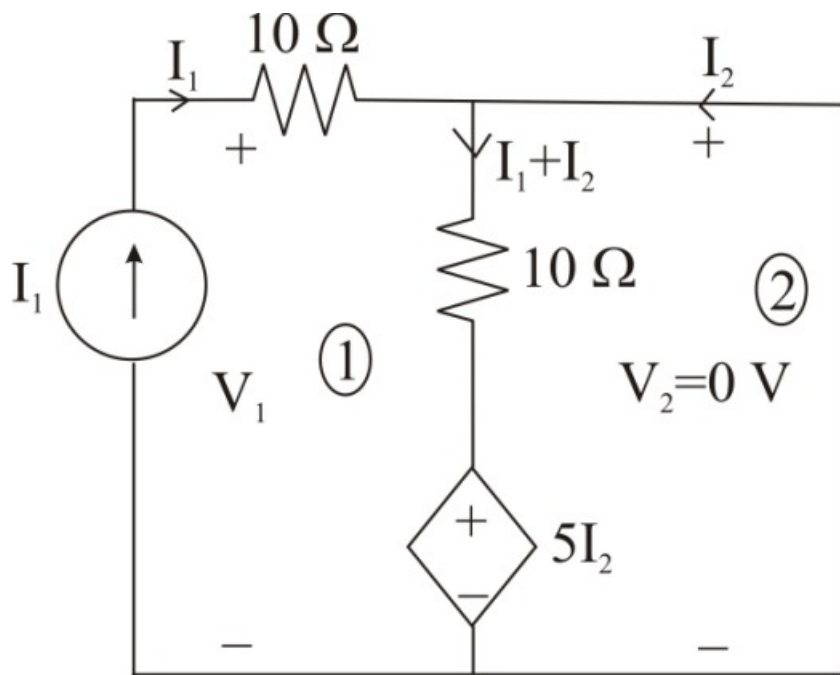


## Step-by-step solution

## Step 1 of 12

Refer to Figure 19.81 in the text book.

Connect the current source  $I_1$  as shown in Figure 1. Identify the currents and voltages in the circuit.



**Figure 1**

## Step 2 of 12

Consider the mesh equation in loop-1.

$$\mathbf{I}_1(10) + (\mathbf{I}_1 + \mathbf{I}_2)(10) + 5\mathbf{I}_2 = 0$$

$$\mathbf{I}_1(10+10) + \mathbf{I}_2(10+5) = 0$$

$$\mathbf{I}_1(20) + \mathbf{I}_2(15) = 0$$

$$\mathbf{I}_1(4) = -\mathbf{I}_2(3)$$

$$\mathbf{I}_1 = -\mathbf{I}_2\left(\frac{3}{4}\right) \dots\dots (1)$$

Therefore, the expression for the current  $\mathbf{I}_1$  is  $-\mathbf{I}_2\left(\frac{3}{4}\right)$ .

### Step 3 of 12

Consider the mesh equation in loop-2.

$$(\mathbf{I}_1 + \mathbf{I}_2)(10) + 5\mathbf{I}_2 = 0$$

$$\mathbf{I}_1(10) + \mathbf{I}_2(10+5) = 0$$

$$\mathbf{I}_1(10) + \mathbf{I}_2(15) = 0$$

$$\mathbf{I}_1(2) + \mathbf{I}_2(3) = 0$$

Substitute  $-\mathbf{I}_2\left(\frac{3}{4}\right)$  for  $\mathbf{I}_1$ .

$$-\mathbf{I}_2\left(\frac{3}{4}\right)(2) + \mathbf{I}_2(3) = 0$$

$$-\mathbf{I}_2(1.5) + \mathbf{I}_2(3) = 0$$

$$\mathbf{I}_2(1.5) = 0$$

$$\mathbf{I}_2 = 0 \text{ A}$$

Therefore, the value of the current  $\mathbf{I}_2$  is 0 A.

### Step 4 of 12

Recall equation (1).

$$\mathbf{I}_1 = -\mathbf{I}_2\left(\frac{3}{4}\right)$$

Substitute 0 A for  $\mathbf{I}_2$ .

$$\begin{aligned}\mathbf{I}_1 &= -(0)\left(\frac{3}{4}\right) \\ &= 0 \text{ A}\end{aligned}$$

Therefore, the value of the current  $\mathbf{I}_1$  is 0 A.

### Step 5 of 12

Determine the expression for the short circuit input admittance  $y_{11}$ .

$$y_{11} = \frac{I_1}{V_1}$$

Substitute 0 A for  $I_1$ .

$$\begin{aligned} Y_{11} &= \frac{0}{V_1} \\ &= 0 \text{ S} \end{aligned}$$

Therefore, the expression for the short circuit input admittance  $y_{11}$  is 0 S.

**Step 6** of 12

Determine the value of the short circuit transfer admittance  $y_{21}$  from port-2 to port-1.

$$y_{21} = \frac{I_2}{V_1}$$

Substitute 0 S for  $I_2$

$$\begin{aligned} y_{21} &= \frac{0}{V_1} \\ &= 0 \text{ S} \end{aligned}$$

Therefore, the value of the short circuit transfer admittance  $y_{21}$  from port-2 to port-1 is 0 S.

**Step 7** of 12

Connect the current source  $I_2$  as shown in Figure 2. Identify the currents and voltages in the circuit.

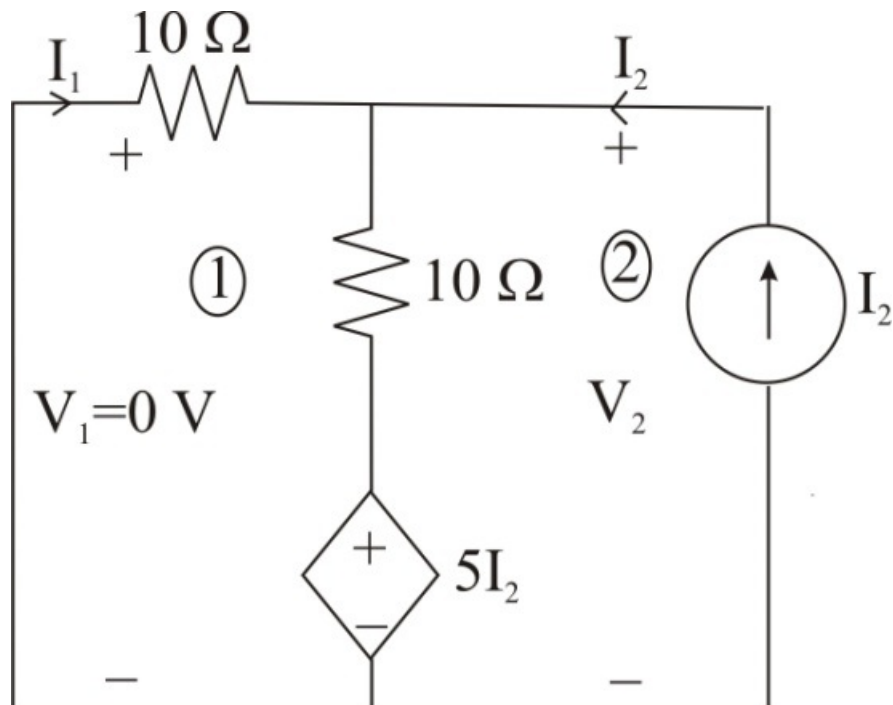


Figure 2

**Step 8** of 12

Consider the mesh equation in loop-1.

$$\mathbf{I}_1(10) + (\mathbf{I}_1 + \mathbf{I}_2)(10) + 5\mathbf{I}_2 = 0$$

$$\mathbf{I}_1(10+10) + \mathbf{I}_2(10+5) = 0$$

$$\mathbf{I}_1(20) + \mathbf{I}_2(15) = 0$$

$$\mathbf{I}_1(4) = -\mathbf{I}_2(3)$$

$$\mathbf{I}_1 = -\mathbf{I}_2\left(\frac{3}{4}\right) \dots\dots (2)$$

Therefore, the expression for the current  $\mathbf{I}_1$  is  $-\mathbf{I}_2\left(\frac{3}{4}\right)$ .

**Step 9** of 12

Consider the mesh equation in loop-2.

$$(\mathbf{I}_1 + \mathbf{I}_2)(10) + 5\mathbf{I}_2 = 0$$

$$\mathbf{I}_1(10) + \mathbf{I}_2(10+5) = 0$$

$$\mathbf{I}_1(10) + \mathbf{I}_2(15) = 0$$

$$\mathbf{I}_1(2) + \mathbf{I}_2(3) = 0$$

Substitute  $-\mathbf{I}_2\left(\frac{3}{4}\right)$  for  $\mathbf{I}_1$ .

$$-\mathbf{I}_2\left(\frac{3}{4}\right)(2) + \mathbf{I}_2(3) = 0$$

$$-\mathbf{I}_2(1.5) + \mathbf{I}_2(3) = 0$$

$$\mathbf{I}_2(1.5) = 0$$

$$\mathbf{I}_2 = 0 \text{ A}$$

Therefore, the value of the current  $\mathbf{I}_2$  is 0 A.

**Step 10** of 12

Recall equation (1).

$$\mathbf{I}_1 = -\mathbf{I}_2\left(\frac{3}{4}\right)$$

Substitute 0 A for  $\mathbf{I}_2$ .

$$\begin{aligned}\mathbf{I}_1 &= -(0)\left(\frac{3}{4}\right) \\ &= 0 \text{ A}\end{aligned}$$

Therefore, the value of the current  $\mathbf{I}_1$  is 0 A.

**Step 11** of 12

Determine the value of the short circuit transfer admittance  $y_{12}$  from port-1 to port-2.

$$y_{12} = \frac{I_1}{V_2}$$

Substitute 0 S for  $I_1$ .

$$\begin{aligned} y_{12} &= \frac{0}{V_2} \\ &= 0 \text{ S} \end{aligned}$$

Therefore, the value of the short circuit transfer admittance  $y_{12}$  from port-1 to port-2 is 0 S.

**Step 12** of 12

Determine the expression for the short circuit output admittance  $y_{22}$ .

$$y_{22} = \frac{I_2}{V_2}$$

Substitute 0 S for  $I_2$ .

$$\begin{aligned} y_{22} &= \frac{0}{V_2} \\ &= 0 \text{ S} \end{aligned}$$

Therefore, the expression for the short circuit output admittance  $y_{22}$  is 0 S.